Section 1. Basic Decomposition of Risk

[문제상황]

data $\{(x_i, y_i)\}_{i=1,\dots,n}$ $\Rightarrow$ map $f: \mathcal{X} \to \mathcal{Y}$

$\begin{cases} \text{i.i.d} \\ x_i \in \mathcal{X} \text{ (high dim)} \\ y_i \in \mathcal{Y} \text{ (label)} \end{cases}$

loss $f: \mathcal{Y} \times \mathcal{Y} \to \mathcal{Y}$

(pointwise)

[Generalization]

(population risk) $R(f) = \mathbb{E}_{(x,y)}[\ell(f(x), y)] = E[\hat{R}(f)]$

(empirical risk) $\hat{R}(f) = \frac{1}{n}\sum_{i=1}^{n} \ell(f(x_i), y_i)$

[Approximation]

- Hypothesis Space $\mathcal{F} = \{f: \mathcal{X} \to \mathcal{Y}\}$ (suppose $\mathcal{F}$ normed space)
- complexity measure $\gamma(f)$, $\|f\|$

① Euclidean Norm of weights

② # of params

③ # of G.D iterations

$\Rightarrow$ let $\mathcal{F}_\delta = \{f \in \mathcal{F} : \gamma(f) \le \delta\}$

ERM (empirical risk minimization)

① $\min_{f \in \mathcal{F}_\delta} \hat{R}(f)$ risk

② $\min_{f \in \mathcal{F}} \hat{R}(f) + \lambda \gamma(f)$ risk complexity

empirical risk ⇓

with limited complexity

③ $\min \gamma(f)$ complexity

when $\hat{R}(f) = 0$ $(f(x_i) = y_i)$

Basic Decomposition of risk

$$\mathrm{I}\quad R(\hat{f}) - \hat{R}(\hat{f}) \quad + \quad \hat{R}(\hat{f}) - \inf_{f \in F_S} \hat{R}(f)$$

optimization error

$$\mathrm{II} \quad + \quad \inf_{f \in F_S} \hat{R}(f) - \inf_{f \in F_S} R(f) \quad + \quad \inf_{f \in F_S} R(f) - \inf_{f \in F} R(f)$$

approximation error

generalization error

$$\leq \quad 2 \cdot \sup_{f \in F_S} \left| R(f) - \hat{R}(f) \right| \quad + \quad \text{opt} \quad + \quad \text{apx}$$

Note. II

$$\boxed{\left| \inf_{F_S} \hat{R}(f) - \inf_{F_S} R(f) \right|} \leq \sup_{F_S} \left| R(f) - \hat{R}(f) \right|$$

(pf)

$$\inf_{F_S} \hat{R}(f) - \inf_{F_S} R(f)$$

$$= \inf_{F_S} \hat{R}(f) - R(f^*)$$

$$\leq \hat{R}(f^*) - R(f^*)$$

$$\leq \text{RHS}$$

$$\inf_{F_S} R(f) - \inf_{F_S} \hat{R}(f)$$

$$= \inf_{F_S} R(f) - \hat{R}(\hat{f})$$

$$\leq R(\hat{f}) - \hat{R}(\hat{f})$$

$$\leq \text{RHS}$$

Section 2. Generalization Error and the Curse of dimensionality

---

trade-off: $\delta \Uparrow$ $n \Downarrow$ then generalization error $\Uparrow$ vs then $\delta \Downarrow$ approximation error $\Uparrow$

---

idea ① $n \Uparrow$ then $\mathcal{E}_{gen} \Downarrow$

$$\frac{1}{n}\sum_{i=1}^{n} \ell(f(x_i), y_i) - \mathbb{E}_{(x,y)}\left[\cancel{\ell(f(x), y)}\right]$$

fixed

empirical risk — population risk

$$\mathrm{Var}\left[\hat{R}(f) - R(f)\right] = \frac{1}{n} \cdot \mathrm{Var}\left[\cancel{\ell(f(x_i), y_i)}\right]$$

fixed.

generalization error — 고정. $n \nearrow$

---

idea ② $d \Uparrow$ then $n \Uparrow$ (to keep $\mathcal{E}_{gen}$ stable)

let $\{(x_i, f^*(x_i))\}_{i=1,\dots,n}$

x_i: $N(0, I_d) \in \mathbb{R}^d$

$f^*(x_i)$: target function

· minimize $\mathbb{E}\left|\hat{f}(x) - f^*(x)\right| \le \varepsilon$

- suppose $f^*(x) = \langle x, \theta^* \rangle$
- then $\mathcal{F} = \{f : f(x) = \langle x, \theta \rangle\} \approx \mathbb{R}^d$
- $n = d$.

idea ③ curse of dimensity : $d \Uparrow$ then $n \Uparrow\Uparrow\Uparrow$ (to keep Egen stable)

- f^* only locally linear & β-Lipschitz.

- $\|f\| = Lip(f) + \|f\|_\infty$

  where $Lip(f) = \inf\{\beta' : |f(x) - f(x')| \le \beta' \|x - x'\|\}$

- $\hat{f} = \underset{f \in F}{\operatorname{argmin}} \{ Lip(f) : f(x_i) = f^*(x_i) \}$

- $|\hat{f}(x) - f^*(x)| \le |\hat{f}(x) - \hat{f}(x_i)| \le Lip(\hat{f}) \|x - x_i\|$

  $+ \cancel{|\hat{f}(x_i) - f^*(x_i)|}$

  $+ |f^*(x_i) - f^*(x)| \le \beta \|x - x_i\|$

- hence, $(MSE) = E|\hat{f}(x) - f^*(x)|^2 \le 4\beta \|x - x_i\|^2$

  $\sim W_2^2(V, \hat{V}_n)$

  $\sim \frac{1}{n^{1/d}}$

- if $\|\hat{f} - f^*\|^2 \le \varepsilon$

  then $n \sim \underbrace{\left(\frac{1}{\varepsilon}\right)}_{\text{아주 큰 상수}}^{d}$

Section 3. Universal Approximation Thm.

UAT : $G(x) = \sum_{j=1}^{N} \alpha_j \, \sigma(w_j^T x + \theta_j)$

are dense in $C(I_n)$

σ : arbitary continuous sigmoid func.

$w_j \in \mathbb{R}^n$, $\alpha_j, \theta_j \in \mathbb{R}$

Note.

$\sigma : \mathbb{R} \to [0, 1]$

is called sigmoid

if ① $\sigma \in C'(\mathbb{R})$

② $\sigma(+\infty) = 1$

③ $\sigma(-\infty) = 0$

$f(x)$: raw system

μ : belief system

$d\mu(x) = \mu(dx)$: weight / importance

$\int f(x)\, d\mu(x)$: final evaluation

σ is discriminatory

for measure $\mu \in M(I_n)$

: $\int_{I_n} \sigma(w^T x + \theta)\, d\mu(x) = 0$

$\forall w \in \mathbb{R}^n$ $\forall \theta \in \mathbb{R}$ then $\mu = 0$

Lemma 1. any conti sigmoid

func is discriminatory for $\forall \mu \in M(I_n)$

Lemma 2. ① U : linear subspace of

normed linear space X.

② U is not dense : $\exists x_0 \in X$ s.t $d(x_0, U) \geq \delta > 0$

then 有 bounded

functional L on X

s.t
① $L|_U = 0$ i.e $L(u) = 0$ $\forall u \in U$

② $L(x_0) = \delta$

③ $\|L\| \leq 1$

$L \neq 0$

Note. bounded linear function

$$\|L\| = \sup_{x \neq 0} \frac{|L(x)|}{\|x\|}$$

Thm. RRT

- compact $K \subseteq \mathbb{R}^n$
- $C(K)$: set of real-valued conti func on K.
- F : bounded linear func on $C(K)$

$\Rightarrow$ 有 unique finite signed measure μ on K

s.t $\underbrace{F(f)}_{\text{linear}} = \int_K f(x)\, \underbrace{d\mu(x)}_{\text{measure}}$ $\forall f \in C(K)$

Lemma 3.

U : linear, non-dense subspace of $C(I_n)$

$\Rightarrow$ 有 $\mu \in M(I_n)$ s.t $\int_U h\, d\mu = 0$ $\forall h \in U$

$\mu \equiv L$ with $L|_U = 0$, $L \neq 0$

(pf) 有 bounded linear function L s.t $L|_U = 0$, $L \neq 0$ on $C(I_n)$ (lemma 2)

有 $\mu \in M(I_n)$ s.t $L(f) = \int_{I_n} h\, d\mu(x)$ $\forall h \in C(I_n)$ (RRT)

$L|_U = 0$: $L(f) = 0$ $(\forall h \in U)$

$L \neq 0$: $\mu \neq 0$

Lemma 4.

σ be any conti. discriminating func.

then $U = \{ G : G(x) = \sum_{j=1}^{N} \alpha_j \cdot \sigma(w_j^T x + \theta_j) \}$

are dense in $C(I_n)$ $(w_j \in \mathbb{R}^d,\ \alpha_j, \theta_j \in \mathbb{R})$

(pf) suppose U not dense

· 有 measure $\mu \neq 0$ s.t $\int_{I_n} h \, d\mu = 0$ $\forall h \in U$

$\downarrow$

$$\sum_{j=0}^{N} \alpha_j \int_{I_n} \sigma(w_j^T x + \theta_j) \, d\mu = 0 \qquad \forall w_j \in \mathbb{R}^n, \ \alpha_j, \theta_j \in \mathbb{R}$$

$\downarrow$

$$\int_{I_n} \sigma(w^T x + \theta) \, d\mu = 0 \qquad \forall w \in \mathbb{R}^n, \ \theta \in \mathbb{R}$$

since σ discriminatory, $\mu = 0$ $\longrightarrow$ contradict.

U is dense

$\Rightarrow$ $\forall f \in C(I_n)$ and $\forall \varepsilon > 0$

有 $G(x)$ s.t $\underbrace{|G(x) - f(x)| < \varepsilon}_{L_\infty \text{ norm}}$ $(\forall x \in I_n)$